

A Deep Learning Approach to Depression Detection Based on Convolution Neural Networks and Transfer Learning

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Abstract—Depression and mental health difficulties have become a big global issue in recent years. It is characterized by a depressed mood and reluctance to action. It is a frequent mental health problem that has been connected to an elevated chance of dying at an early age. Mental health is a subject that most families do not wish to pay as much attention to as it should, especially in our country, and as a result, many persons suffering from Major Depressive Disorder (MDD) are often vulnerable. For this reason, acknowledging this situation as well as detecting it as early as possible is very important. At present there are numerous ways to detect depression by various methods such as emotion recognition, daily routine analysis etc. But there is lack of research about which particular model is better to work with in this case and the reason behind this. In our research we have used the EEG data of several healthy and MDD patients. This research seeks to diagnose depression using EEG brain waves and multiple deep learning/transfer learning models (for example, VGG16, Xception, ResNet152, MobileNetV2 etc.) to find the model that delivers the highest level of accuracy for our data type. Also we want to research why a certain model performs better and what factors might influence model performance in order to suggest a model that can be used in the majority of cases for identifying depression and model improvement. Moreover, we created our own customized model that provides the highest level of accuracy (99.75%). We have succeeded in achieving the highest level of accuracy among the existing models that we have applied. We hope that this study will contribute in the development of novel ways for developing models and detecting depression at the shortest possible time.

Index Terms—Major depressive disorder, Emotion recognition, Brain waves, Transfer learning model, VGG16, Xception, ResNet152, MobileNetV2.

I. INTRODUCTION

DEPRESSION is a type of mood illness characterized by a continuous sense of melancholy and a loss of interest. It creates a huge problem in a person's day to day life. Around 280 million people worldwide suffer from depression [1]. Depression is distinct from normal mood swings and short-term emotional responses to ordinary difficulties. It can be dangerous to one's health, especially if it is persistent and has a moderate or severe intensity. It can cause a lot of pain and make it difficult for the person to function. Depression can lead to suicide in the worst-case scenario. Every year, around 700,000 people die by suicide. Despite the fact that there are effective treatments for mental diseases, more than 75% of people in low and middle income nations do not

obtain them [2]. People with depression are frequently not given the proper treatment and administered antidepressants in countries of all income levels. Many deep learning approaches are currently available that can detect depression based on many sorts of inputs such as EEG graphs, personal history, social media history, and so on. Different sorts of data, on the other hand, necessitate different types of models. There are numerous approaches to detect depression. But there is no universal procedure or guideline that can assist us in creating a customized model or modifying an existing model. Furthermore, the ability to detect depression as soon as feasible is critical. Otherwise, it will be quite difficult for the patient to begin and maintain the necessary therapy. Various studies have been conducted on this topic to date, but in our research, we will do a thorough study based on previous research and develop our own model.

Much research has been done and developed over the past few years on depression analysis through EEG signals. A study regarding identification stress using EEG signals [3] collected their dataset from 86 automotive drivers and classify driving-induced stress patterns using machine learning-based approach and EEG signals. They performed three methods Neural Network (NN), SVM and Random Forest (RF) to break down the patterns of EEG on the basis of the self-assessment of the emotional state of the tested person when they drove in various situations. Tested over 50 automotive drivers, SVM performed better along with $97.95\% \pm 2.65\%$ accuracy as well as 88.83% sensitivity and 94.92% specificity. Research was done on [4] electroencephalogram based MDD diagnostic models using an advanced computational neurobiology approach combined using a deep convolutional neural model. They used ResNet50, MobileNet and Inceptionv3 to separate MDD subjects from normal persons. EEG dataset used in the study were collected from 46 control persons and 46 depressed patients, approved by a local medical research committee. They found that MobileNet architecture provides classification accuracy of 89.33% and 92.66%. In terms of bandwidth, delta band outperforms with a prediction accuracy of 90.22% and area under curve showed 0.9 for ResNet50 architecture.

Similarly, the authors [5] offered a multimodal fusion of speech and language presentation to detect depression in their work. They trained the model to derive subject scores on the Patient Health Questionnaire (PHQ) and used deep spectral characteristics extracted from a previously trained VGG16 net-

work as well as Gated Convolutional Neural Network (GCNN) accompanied by LSTM for speech method. They obtained CCC scores of 0.497 and 0.608 in EDAIC. In addition, they combined the two methods using feature fusion, applying the final representation of each model from a separate method to a fully connected layer to evaluate. By the use of a multimodal approach, the CCC value of the development set is 0.696 and the CCC value of the EDAIC corpus is 0.403, which shows an absolute improvement of 0.283 points compared with the initial value. A study on EEG based depression detection [6] used KNN, SVM, CT, and ANN (Artificial Neural networks); these four algorithms on 250 participants were 92 depressed patients and 121 normal subjects in total. All of these classifications and 10-fold cross-validation were performed using the MATLAB software version of R2014a. Among all selected algorithms, the performance of KNN was best and the accuracy rate was 79.27%. The outcomes likewise showed featuring “absolute power of theta wave” to be processed with a quicker data handling productivity. The authors [7] used some classification models such as SVM, logistic regression (LR), and naive Bayesian (NB) in their work. This model employed the relationship between EEG functions and study groups between MDD patients and healthy persons. The dataset was collected from 34 MDD patients with 18 women (age, mean = 40.33, standard deviation $\pm$ 12 861) and a group of 30 healthy persons with women = 9 of the same age (age, mean = 38 237, standard deviation $\pm$ 15, sixty-four). The results show that the odds are better than the chance. They found that the study results SVM performed with accuracy of 98%, 99.9% sensitivity and 95% specificity. Whereas LR performed accuracy of 91.7%, 86.66% sensitivity and 96.6% specificity. Lastly NB performed accuracy of 93.6%, 100% sensitivity and specificity of 87.9. The authors finally concluded by saying that Synchronization likelihood (SL) features may be a promising diagnostic tool for depression. A research [8] was done on cognitive depression detection using EEG where they collected EEG data of 80 subjects and out of these, 76 subjects were classified. They collected their dataset from PhysioNet and PhysioBank where every dataset accommodates one-minute EEG data. They used DFT, FFT on the occipito-parietal wave which was produced from occipital as well as parietal parts of their brain. In their research, 37.5% of subjects showed signs of MDD whereas 80% of subjects showed a small amount of depression and last 20% of the subjects did not show statistical EEG parameters that would indicate depression. In addition, they could not classify 2.5% of the subjects because of some noises in the EEG signal. This research is about a comparative study. With our dataset, we have run different models to see which one has the best accuracy at diagnosing depression. Then we are going to look at the model layer by layer to see what might be causing it to work better than others. Then we’ll flip the layers to examine the consequences of each adjustment or which combination gives the best results. Furthermore, we have come up with our own customized model which gives the highest accuracy. As previously stated, there are numerous deep learning models for detecting depression. For improved accuracy, we’ve proposed a method and a model for creating

customized or modified models. Our hypothesis demonstrates how adding or removing layers from our modified model alters the outcomes. This will aid researchers in developing better models and comparing their results in the future. Moreover, our model demonstrates optimal performance. Now that we’ve established a new model, comparative analysis will be more useful if someone wants to work with our datasets. Early detection of depression will be easier, according to our research.

II. PROPOSED METHOD

We used 2D CNN along with nine other deep learning models like VGG16, ResNet152, Xception, InceptionV3 etc. to execute tests on our data.

A. Data Collection and Data Pre-Processing

The dataset “MDD Patients and Healthy Controls EEG Dataset” contains 193 EEG signals from two different types of individuals, as in Signals are classified into one of two categories: “MDD,” which refers for “Major Depressive Disorder,” and “Healthy,” which stands for “Healthy Controls.” Furthermore, each of the categories has its own set of sub-categories. The letters “EO” stand for “Eyes Open,” “EC” for “Eyes Closed,” and “Task” for “P300 data,” which means “event-related potential (ERP) endogenous.” Moreover, we obtained this data through “Figshare” [9]. Moreover, the EEG signals in the dataset contain some properties like Channels (20/22 channels), Sampling Frequency (256.0 Hz), Data Shape etc. From Table I, the dataset information is clearly visible:

	MDD = Major Depressive Disorder	H = Healthy Controls
EO	32	30
EC	30	28
Task	33	28

TABLE I
DATASET TABLE FOR NUMBER OF SAMPLE PATIENTS

Our dataset’s data type was “EDF” format which is known as “European Data Format”. We analyzed some data and came to know that there are 22 overall channels in our brain where researchers collect the EEG signal. But in some cases, we noticed the number of channels available on our data is 20. The total duration of each data which varies from five minutes to ten minutes and sampling frequency. So, in a few cases a couple of channels were left out. Here is a visual version (Figure 1) or those areas in our brain:

B. Dataset Pre-Processing

After importing our dataset, we need to label each of our data. EEG signals are generally displayed as a graph where “Brain Channel” is represented as “Rows” and “Time” is represented as “Columns”. We run every single data one by one and from there we get two corresponding graphs for each data we have. We converted each graph as an image and saved it for our implementation phase. Each image is labeled as the respective unique subject code. For example, the loaded EEG graphs for data “MDD EO 17” means the graphs of this EEG

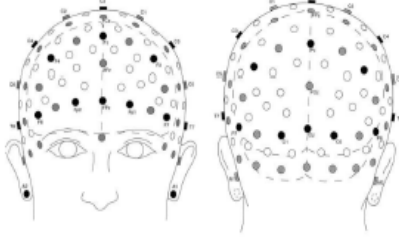


Fig. 1. EEG Signal Locations

data are saved as two images which are labeled as “MDD EO 17 (1)” and “MDD EO 17 (2)”. The program will iterate through all data saved in different folders and do the same thing for all the data we have. This below figure (Figure 2) is an example:

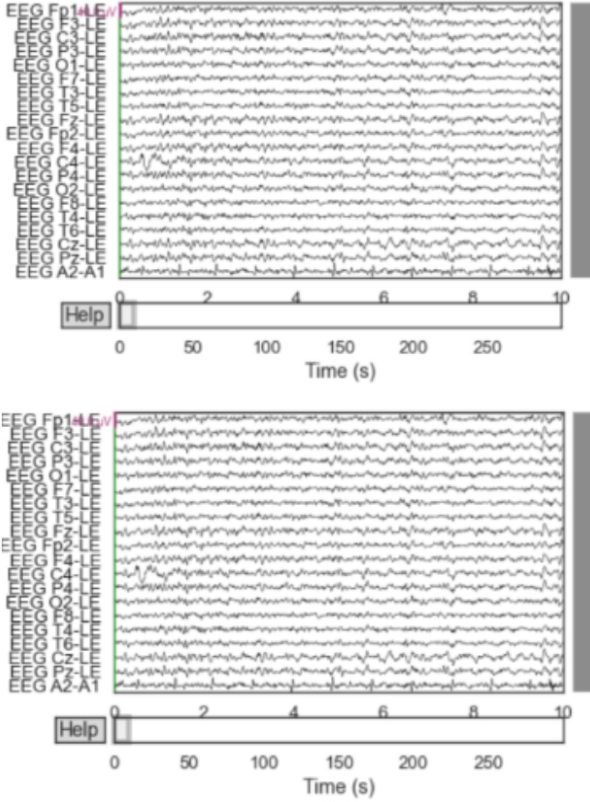


Fig. 2. EEG Signal Graph

Training a convolutional neural network on raw images, for example, is likely to result in poor classification. Before being passed to the following ResNet152, ResNet152V2, VGG16, VGG19, InceptionResNetV2, InceptionV3, Xception, DenseNet169, DenseNet201, MobileNetV2, 2D CNN (with and without Augmentation) module, this study ran through several pre-processing. They are data resizing, data reshaping, data augmentation, data classification etc. Here we basically classified our dataset into two types of sets. Firstly, for the 2D CNN model we created two classes named “Healthy” and

“MDD”. And for other models of convolutional layers, we further classified our dataset into “Train” and “Test” for each of the two mentioned classes “Healthy” and “MDD”.

The figure 3 shows the data processing on our EEG dataset:

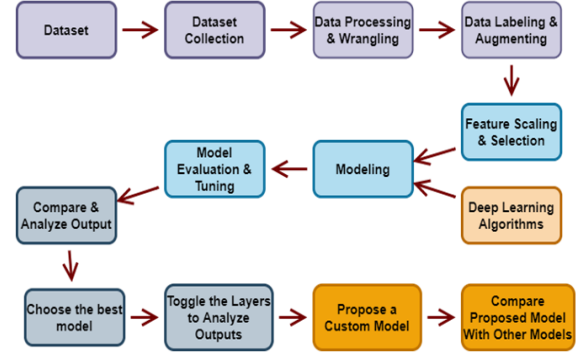


Fig. 3. Data Collection Process

C. Models

We divided the whole dataset into two parts. We used 80% of our data from both “Healthy” and “MDD” for training the model and 20% for testing the model.

In this research, we tried to detect depression using 2D CNN and some other deep learning models. For the experiment, we used nine deep learning models namely VGG16, VGG19, Xception, InceptionV3, ResNet152, InceptionResNetV2, MobileNetV2, DenseNet169 and DenseNet201.

In this research paper, we compare and analyze depression detection with different transfer learning models. We use a small digital network, transmit it to the image and adjust it according to the channel value.

Convolutional neural network (CNN) is a type of deep learning method that has recently gained popularity in deep neural networks. It uses input images and teaches the meaning of different objects (learning weights and distortions), assigning pictures and being able to distinguish one from the other.

Our 2D CNN model had pooling and fully connected layers and we used different activation functions like Sigmoid function, Hyperbolic Tangent function (TanH) and Rectified Linear unit function (ReLU) where ReLU worked best by not having backpropagation error and avoiding gradient vanishing.

The sigmoid function is avoided in more complex and advanced neural models. The equation (1) of the sigmoid function is given:

$$\text{Sigmoid}(x) = \frac{1}{1 + e^{-x}} \quad (1)$$

Here, it exists between (0,1). Therefore, when the probability of predicting is between 0 and 1, the sigmoid chart is used. It has a differentiable S-curve. This is a monotonic function. The biggest disadvantage of this feature is data loss, which increases as the network grows deeper. For this reason, we used another function called Hyperbolic Tangent Function (TanH) to assign classes.

$$y = 2\sigma(2x) - 1 \quad (2)$$

This function has a central zero point, but there is another issue called Vanishing Gradient issue. To avoid all the problems, we used Rectified Linear Function (ReLU) which is a piecewise linear function that outputs directly from an input. It has a range $(0, \infty)$. In the realm of deep learning, it is the most well-known activation function and can be summarized by the following equation (3):

$$y = \max(0, x) \quad (3)$$

The core advantage of using ReLU is that it fixes the vanishing gradient problem, compared with TanH, it has low activation (50%) and no backpropagation error. The disadvantage of ReLU is that this function has no zero center and cannot be differentiated to 0. Another drawback is the dying ReLU problem, which occurs when half of the ReLU's output is dormant and used for non-zero center actions.

Furthermore, leaky ReLU was used in case of any negative values, softmax and fine tuning was done in order to get desired results.

In our research we used Transfer Learning models to train our CNN model. The layers of the learned model are then used to train a new model on the problem of interest. Dense Convolutional Network, commonly known as DenseNet, is a feed-forward network that pre-connects each layer to all other layers. DenseNet 201 is another image classification model from the DenseNet family. It is a 201-layer deep convolutional neural network. Densenet169 is a typical output of an object classifier for 1000 different classifiers that fit the ImageNet database and has 169 layers. The key difference between DenseNet 169 and this model is its size. In addition, we used InceptionV3 which has 42 layers. Another deep learning model was used here is InceptionResNetV2, this model is a convolutional neural network that was trained on over a million images from the ImageNet database. It has 164 layers and can categorize photos into 1000 different items.

MobileNetV2 architecture contains a total of 32 convolutional layers with 32 filters, followed by the remaining 19 bottleneck layers [10]. It is 53 layers deep and has learned multifunctional representations for many types of images with an input size of 224-by-224 [11]. Residual networks (ResNET) are a group of deep neural networks that have similar topologies but differ in depth [12]. We used ResNet152 which has 152 layers. Other models are VGG16 and VGG19. VGG16 is a convolutional neural network model, The input to conv1 layer is of fixed size 224 x 224 RGB image with 23 layers. Whereas, VGG19 is a VGG variation of 26 layers containing 16 convolutional layers and 3 fully connected layers with 5 MaxPool layers and 1 SoftMax layer and 16 convolutional layers [13]. Compared to VGG16, VGG19 is slightly better, but it requires more memory. There are 16 layers in total with 5 blocks and each block has a maximum pooling layer [14]. Lastly, Xception is a 71-layer deep convolutional neural network with an input image size of 299 x 299 [15].

Lastly, Two or more things or concepts are investigated and compared in a comparative study. Investigations, comparisons, and contrasts of subjects or concepts are all part of comparative studies. In this section, we will discuss the results of our used models and compare them with our proposed model.

III. EXPERIMENTAL RESULTS AND DISCUSSION

A. Results of Supervised Models

First, we applied a very basic CNN model to our dataset. Initially we applied 2D CNN. Later to improve the accuracy level, we applied data augmentation. This is the result we got from our model:

2D CNN	Value Loss	Accuracy (Highest)
Without Augmentation	0.6082	78%
With Augmentation	0.2741	89%

TABLE II
BASIC CNN MODEL RESULT

Then, we applied our dataset to a few well-known existing deep learning models using Keras Application [16]. Prediction, feature extraction, and fine-tuning are all possible with these models. Weights of each model are automatically downloaded when we initialize the models. Here are the results of our implemented deep learning models (Input size 224x224).

Deep Learning Models	Value Loss	Accuracy (Highest)
DenseNet169	0.9921	91%
DenseNet201	0.8471	93%
InceptionV3	1.6879	91%
InceptionResNetV2	0.9383	89%
MobileNetV2	1.6837	92%
ResNet152	0.5131	78%
VGG16	0.0804	97%
VGG19	0.2148	92%
Xception	1.0458	88%

TABLE III
DEEP LEARNING MODEL RESULT WITH INPUT SIZE = 224x224

The highest accuracy we got is 97% from the model VGG16.

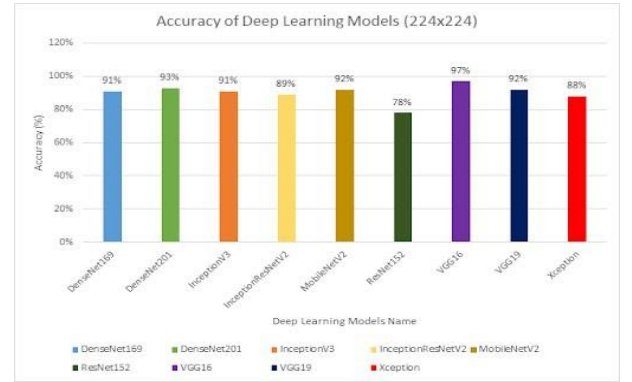


Fig. 4. Deep Learning Model Result

B. Analysis of Comparative Results

Comparing results with previous research and papers is a good way to show how our research has improved or aggravated the task. Wajid Mumtaz originally posted the dataset we used on "Figshare". He and his fellow researchers also used this dataset to conduct a few researches.

In the paper, "A deep learning framework for automatic diagnosis of unipolar de- pression", Wajid Mumtaz and Abdul

Qayyum worked with different portions of the dataset to detect depression [17]. In one attempt, they took 15 depressive and 15 healthy cases and the accuracy they achieved is 93%. This deep learning model has 13 layers CNN. They also used the 1D CNN model here.

Other papers used different deep learning models to detect depression or for some other purposes. We are going to implement CNN models and measure how far we can get. We are going to use a 2D CNN model.

Therefore, the models' accuracy does not depend on any specific variable. We always need to tweak things and change value to monitor when its accuracy is increasing or decreasing. After so many attempts of trial and error, a model can be perfect.

C. Findings While Proposed Customized Model

We tried to propose a new model for this dataset to detect depression. We started by making use of the VGG16 model systematically. As it gave us the highest accuracy but when we ran our dataset to our custom VGG16 model, its accuracy was not up to the mark. It is because in the available built-in model, the model is optimized. Besides, it is difficult to make an optimized model step by step. That was why we did not get the accuracy we got in the VGG16 model.

Then at first, we tried to add more layers to the problem. However, the accuracy declined. We tweaked a few parameters and added some built-in methods for example dropout, batch normalization, optimizer-learning algorithm: AdaM to see how it reacts.

While tweaking the parameters, we observed a few patterns. They are if we increase the number of convolutional layers the accuracy usually decreases. Nonetheless, if we remove a few layers the accuracy increases. Additionally, in case of dropout, the more the dropout rate the lower the accuracy and vice versa.

Other methods like batch normalization and optimizer algorithms do not have that kind of consistency in case of increasing or decreasing. In case of convolutional layers and dropout rate, it always stayed in the flow.

One thing we need to mention here is, the pattern we are seeing in case of layers and dropouts, specifically based on our dataset. There is no warranty that the models will always behave like this. It may vary from model to model and dataset to dataset.

Our hypothesis is, as our dataset is very small if we compare it to other datasets, is thereason behind this.

Finally, after a lot of test runs, we found a model giving the best accuracy among all models. We have considered image of size 224x224 as an input of this CNN model. For proposed 2D-CNN approach, we have customized the dimensions of different layers of the convolutional neural network. In the first layer, the convolutional layer where 16 weights of dimension (3,3) are applied with a rectified linear unit activation function. The max-pool layer has a pool size of 2x2 is which is implemented to the image alongside batch normalization and a dropout rate of 10%. Besides, the second convolutional layer where 32 weights of dimension (3,3) are implemented

including a rectified linear unit activation function. The second max pool layer has a pool size of 2x2 is implemented alongside batch normalization and a again a dropout rate of 10% to the image. Then, the third convolutional layer where 48 weights of dimension (3,3) are implemented including a rectified linear unit activation function. The third max-pool layer has also a pool size of 2x2 is implemented to the image with batch normalization and dropout rate of 10%. Figure 6 is the architecture of our proposed model:

The highest accuracy we got from this model is 99.75% with a value loss 0.0122.

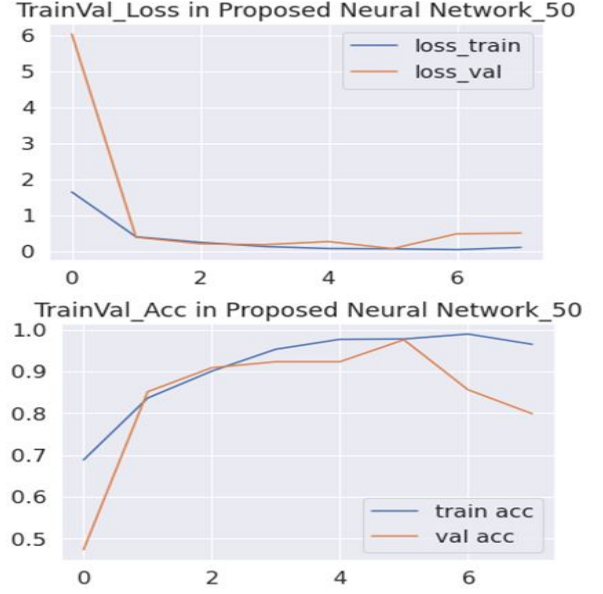


Fig. 5. Graph of Train Value Loss Accuracy

In our research, our proposed deep convolutional neural network model has got 99.75% test accuracy with value loss of 0.0122 for detecting depression based on our dataset. While on the available models and on the previous research based on this dataset wasn't able to achieve this level of accuracy. Also, our test accuracy is more significant if we compare it to the other previously used models in past research.

Epoch	Training Loss	Training Accuracy(%)
1	2.4516	64.94
2	0.4708	76.29
3	0.3663	85.01
4	0.1820	92.63
5	0.0546	97.88
6	0.0319	98.90
7	0.0220	99.32
8	0.0118	99.75

TABLE IV
TRAINING ACCURACY AND TRAINING LOSS OF PROPOSED MODEL

Proposed Convolutional Neural Network Model	
Testing Accuracy (Highest)	Testing Loss
99.75%	0.0122

TABLE V
TESTING ACCURACY AND TESTING LOSS OF PROPOSED CNN MODEL

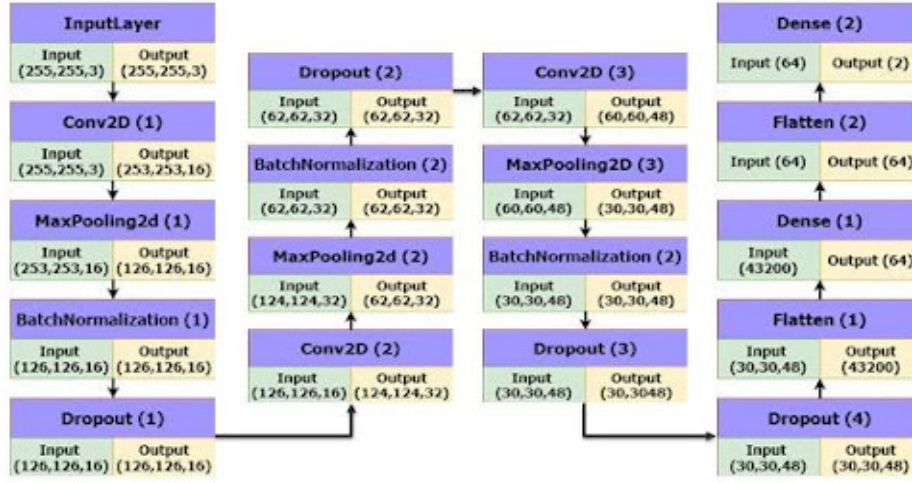


Fig. 6. Architecture of Proposed Model

IV. CONCLUSION AND FUTURE WORK

Depression is more than merely being sad or going through a difficult time. It's a major mental health problem that necessitates knowledge and treatment. Depression, if left untreated, can be devastating to those who suffer from it as well as their family. Many people can and do get better with early identification, diagnosis, and a treatment plan that includes medication, counseling, and good lifestyle choices. But till now there are many barriers to detect depression and taking proper treatment. In our thesis, we used a variety of models to detect depression and determined which one was the most accurate based on the results. Furthermore, we are conducting a comparison analysis of the many models we have implemented, based on previous work with these models. Our goal is to figure out which one performs better and why. As a result, it will be useful for anyone who wants to do future study on this topic by allowing them to select the most accurate and efficient model. Furthermore, it can aid in the early detection of depression in MDD patients. Our future goal is to implement an idea of effective treatment for the MDD patients. Our concept is to create a chip for an MDD patient's brain. So, when a depressed patient is experiencing a severe condition, such as suicidal thoughts, panic attacks, or something distressing, the chip will gather a distress signal from the patient and decipher the pattern of the signal as well as the patient's condition. As a result, the chip will alter this signal and send another signal to the patient's neuron, making him joyful or content for the time being and removing the suicidal notion from his head.

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